

**COURSE CSE703095
SOFTWARE REQUIREMENTS**

SAMPLE REPORT - LAB 03

Specification & Documentation

Topic: Illustrative example of a completed report (reference only, do not copy)

Case study project: MedBook – Online Medical Appointment Booking System
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1 Requirements Rewritten in EARS Syntax

ID	Before (original description)	After (EARS syntax)
FR-003	"Book an appointment online"	"When a patient confirms a chosen available time slot, the system SHALL create the appointment with status pending_confirmation within 2 seconds."
FR-004	"Cancel/reschedule an appointment"	"When a patient requests cancellation and at least 2 hours remain, the system SHALL change the appointment status to cancelled."
FR-008	"Drug interaction/allergy warning"	"When a doctor adds a drug to a prescription and it matches a recorded allergy, the system SHALL display a warning before allowing the prescription to be saved."

1.1 Self-assessment Against 6 Quality Attributes (Wiegers)

All 3 rewritten requirements achieve *Verifiable* (clear measurable criteria: 2 seconds, 2 hours) and *Unambiguous* (no vague terms like "quickly" remain). FR-008 in particular needs an added boundary condition (brand-name drugs with different names but the same active ingredient) to fully achieve *Complete*.

2 Auto-generated SRS Document

```

1 [OK] SRS generated: SRS_MedBook_generated.md
2   - 12 functional requirements
3   - 10 non-functional requirements
4   - 8 stakeholder/user groups

```

Listing 1: Console output from srs_generator.py

3 What Automation Handles Well vs. What Needs Human Input

- **Automation handles well:** formatting consistency, no requirement is missed from the dataset, instantly updated when the CSV changes.
- **Needs human input:** business narrative (Sections 1.2, 2.1), supporting diagrams, and especially the Design Constraints section – since constraints usually live in a domain expert’s head, not in a structured CSV.

4 Manually Added Design Constraints

1. The system must be deployed on a cloud platform with data centers located within the country (per data-residency regulation).
2. The UI must support a minimum mobile screen resolution of 360px.
3. All APIs must follow RESTful conventions and return JSON.

5 Extension Code

```

1 def section_glossary(frs, nfrs):
2     import re
3     abbrevs = set()
4     for r in frs + nfrs:
5         text = " ".join(r.values())
6         abbrevs.update(re.findall(r"\b[A-Z]{2,5}\b", text))
7     lines = ["## Auto-generated Glossary\n"]
8     for a in sorted(abbrevs):
9         lines.append(f"- **{a}**: (fill in manually)")
10    return "\n".join(lines)

```

6 Self-assessment

Criterion	Max	Self-score	Notes
Accurate EARS rewrites	2.5	2.5	3/3 requirements meet the standard
Complete auto-generated SRS	3.0	3.0	All 4 sections present
Working extension code	2.5	2.0	-format=latex flag not 100% complete

Criterion	Max	Self-score	Notes
Presentation	2.0	2.0	Follows the template
Total	10.0	9.5	